Sum and Difference Formulas

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Exact values with sum and difference formulas**

Use a sum or difference formula to find the exact value of $\cos 75°$.

$\cos 75° = \cos(45° + 30°) = \cos 45°\cos 30° - \sin 45°\sin 30° = \frac{\sqrt{6}-\sqrt{2}}{4}$.

2 **[Foundational]** • **Target: Exact values with sum and difference formulas**

Find the exact value of $\sin \frac{\pi}{12}$.

$\sin \frac{\pi}{12} = \sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \sin\frac{\pi}{3}\cos\frac{\pi}{4} - \cos\frac{\pi}{3}\sin\frac{\pi}{4} = \frac{\sqrt{6}-\sqrt{2}}{4}$.

3 **[Foundational]** • **Target: Working with given values**

Angles A and B are both in Quadrant I with $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$. Find:

a $\sin(A + B)$ $\cos A = \frac{3}{5}$, $\sin B = \frac{12}{13}$, so $\sin(A + B) = \frac{4}{5} \cdot \frac{5}{13} + \frac{3}{5} \cdot \frac{12}{13} = \frac{56}{65}$.

b $\cos(A + B)$ $\cos(A + B) = \frac{3}{5} \cdot \frac{5}{13} - \frac{4}{5} \cdot \frac{12}{13} = -\frac{33}{65}$.

c the quadrant of $A + B$ Sine positive, cosine negative: Quadrant II.

(a) $\cos A = \frac{3}{5}$, $\sin B = \frac{12}{13}$, so $\sin(A + B) = \frac{4}{5} \cdot \frac{5}{13} + \frac{3}{5} \cdot \frac{12}{13} = \frac{56}{65}$.; (b) $\cos(A + B) = \frac{3}{5} \cdot \frac{5}{13} - \frac{4}{5} \cdot \frac{12}{13} = -\frac{33}{65}$.; (c) Sine positive, cosine negative: Quadrant II.

4 **[Foundational]** • **Target: Working with given values**

Use a sum or difference formula to simplify:

a $\sin\left(x + \frac{\pi}{2}\right)$ $= \cos x$

b $\cos(x - \pi)$ $= -\cos x$

(a) $= \cos x$; (b) $= -\cos x$

5 **[Foundational]** • **Target: Working with given values**

Angles A and B are both in Quadrant II with $\sin A = \frac{8}{17}$ and $\sin B = \frac{3}{5}$. Find $\cos(A - B)$ exactly.

In QII, $\cos A = -\frac{15}{17}$ and $\cos B = -\frac{4}{5}$. $\cos(A - B) = \cos A\cos B + \sin A\sin B = \left(-\frac{15}{17}\right)\left(-\frac{4}{5}\right) + \frac{8}{17} \cdot \frac{3}{5} = \frac{60}{85} + \frac{24}{85} = \frac{84}{85}$.

6 **[Foundational]** • **Target: Verifying identities using sum and difference formulas**

Show that $\tan\left(x + \frac{\pi}{4}\right) = \frac{1 + \tan x}{1 - \tan x}$ using the tangent sum formula $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$.

With $B = \frac{\pi}{4}$, $\tan B = 1$: $\tan\left(x + \frac{\pi}{4}\right) = \frac{\tan x + 1}{1 - \tan x \cdot 1} = \frac{1 + \tan x}{1 - \tan x}$.

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7 [Foundational] • Target: Applications

Two musical tones combine to form a sound wave modeled by $y = \sin\left(x + \frac{\pi}{6}\right)$. Expand this expression using the sum formula, and evaluate the coefficient of $\cos x$ exactly.

$\sin\left(x + \frac{\pi}{6}\right) = \sin x \cos\frac{\pi}{6} + \cos x \sin\frac{\pi}{6} = \frac{\sqrt{3}}{2}\sin x + \frac{1}{2}\cos x$. The coefficient of $\cos x$ is $\frac{1}{2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Additional practice

Angles A and B satisfy $\tan A = \frac{3}{4}$ (QI) and $\tan B = \frac{5}{12}$ (QI). Find $\tan(A + B)$ using the tangent sum formula.

$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{3/4 + 5/12}{1 - (3/4)(5/12)} = \frac{14/12}{1 - 15/48} = \frac{7/6}{33/48} = \frac{7}{6} \cdot \frac{48}{33} = \frac{56}{33}$.

9 [Intermediate] • Target: Additional practice

Simplify $\cos\left(\frac{\pi}{2} - x\right)\cos x + \sin\left(\frac{\pi}{2} - x\right)\sin x$ using the cosine difference formula in reverse.

This matches the pattern $\cos A \cos B + \sin A \sin B = \cos(A - B)$ with $A = \frac{\pi}{2} - x$, $B = x$: $= \cos\left(\frac{\pi}{2} - 2x\right)$.

10 [Intermediate] • Target: A worked derivation

Starting from the cosine difference formula $\cos(A - B) = \cos A \cos B + \sin A \sin B$, derive the cosine sum formula $\cos(A + B)$ by substituting $B \to -B$.

$\cos(A + B) = \cos(A - (-B)) = \cos A \cos(-B) + \sin A \sin(-B)$. Since cosine is even and sine is odd:
$= \cos A \cos B - \sin A \sin B$.

11 [Intermediate] • Target: A worked derivation

Find the exact value of $\sin 15° \cos 15°$ using a sum/difference-related double angle shortcut (recall $\sin(2\theta) = 2\sin\theta\cos\theta$).

$\sin 15° \cos 15° = \frac{1}{2}(2\sin 15° \cos 15°) = \frac{1}{2}\sin 30° = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

12 [Intermediate] • Target: Sum and difference formulas with tables of values

Given $\sin A = \frac{7}{25}$ (QI) and $\cos B = \frac{4}{5}$ (QI), find $\sin(A - B)$ exactly.

$\cos A = \frac{24}{25}$, $\sin B = \frac{3}{5}$. $\sin(A - B) = \sin A \cos B - \cos A \sin B = \frac{7}{25} \cdot \frac{4}{5} - \frac{24}{25} \cdot \frac{3}{5} = \frac{28}{125} - \frac{72}{125} = -\frac{44}{125}$.

13 [Intermediate] • Target: Sum and difference formulas with tables of values

Given $\tan A = 2$ (QI) and $\tan B = 3$ (QI), find $\tan(A - B)$ using the tangent difference formula.

$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} = \frac{2 - 3}{1 + 6} = -\frac{1}{7}$.

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14 [Intermediate] • Target: Expanding a general sinusoidal expression

Expand $\cos\left(x - \frac{\pi}{3}\right)$ fully using the difference formula, leaving the answer in terms of $\sin x$ and $\cos x$.

$\cos\left(x - \frac{\pi}{3}\right) = \cos x \cos\frac{\pi}{3} + \sin x \sin\frac{\pi}{3} = \frac{1}{2}\cos x + \frac{\sqrt{3}}{2}\sin x$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A geometric interpretation • Cross-Lesson: 8.02 Graphing Rational Functions

Explain, using the unit circle, why $\cos(A - B)$ (not $\cos A - \cos B$) correctly gives the cosine of the angle between two points at angles A and B on the unit circle.

The angle between two points on the unit circle at angles A and B is $|A - B|$, and by definition the cosine of that INCLUDED angle is $\cos(A - B)$ — a genuinely different quantity from the difference of the individual cosine values $\cos A - \cos B$, which has no direct geometric meaning as an angle's cosine.

16 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 8.03 Introduction to Limits

Solve $\sin\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$ for $x \in [0, 2\pi)$ by first expanding the left side.

Expand: $\sin x \cos\frac{\pi}{6} + \cos x \sin\frac{\pi}{6} = \frac{1}{2} \Rightarrow \frac{\sqrt{3}}{2}\sin x + \frac{1}{2}\cos x = \frac{1}{2}$. Alternatively, recognizing $\sin(\theta) = \frac{1}{2}$ directly: $x + \frac{\pi}{6} = \frac{\pi}{6}$ or $x + \frac{\pi}{6} = \frac{5\pi}{6}$, giving $x = 0$ or $x = \frac{2\pi}{3}$.

17 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Given that a signal is modeled by $y = \cos\left(3x - \frac{\pi}{4}\right)$, expand it using the difference formula and identify the coefficients of $\cos 3x$ and $\sin 3x$.

$\cos\left(3x - \frac{\pi}{4}\right) = \cos 3x \cos\frac{\pi}{4} + \sin 3x \sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}\cos 3x + \frac{\sqrt{2}}{2}\sin 3x$. Coefficients: both $\frac{\sqrt{2}}{2}$.

18 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 9.02 Geometric Sequences and Series

Two AC voltage signals are given by $V_1(t) = \sin(\omega t)$ and $V_2(t) = \sin\left(\omega t + \frac{2\pi}{3}\right)$ (phase-shifted by $120°$, common in three-phase power systems). Expand $V_2(t)$ fully using the sum formula, in terms of $\sin(\omega t)$ and $\cos(\omega t)$.

$V_2(t) = \sin(\omega t)\cos\frac{2\pi}{3} + \cos(\omega t)\sin\frac{2\pi}{3} = -\frac{1}{2}\sin(\omega t) + \frac{\sqrt{3}}{2}\cos(\omega t)$.

19 [Challenging] • Target: A geometric interpretation • Cross-Lesson: 9.03 The Binomial Theorem

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Step 1: Compute $\cos\theta$ via Pythagorean identity: $\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$. Step 2: Compute $\tan\theta$: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$. Step 3: Compute $\sin(2\theta)$ using the double-angle identity: $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.

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20 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 1.01 Functions and Function Notation

A radioactive substance follows the decay model $M(t) = M_0 e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

If 20% decays in 15 days, then 80% remains at $t = 15$: $0.80M_0 = M_0 e^{-15k} \Rightarrow e^{-15k} = 0.80$. Take the natural logarithm of both sides: $-15k = \ln(0.80) \Rightarrow k = -\frac{\ln(0.80)}{15} = \frac{\ln(1.25)}{15} \approx 0.01488$ day^{-1}. The half-life $t_{1/2}$ satisfies $e^{-kt_{1/2}} = 0.5$: $t_{1/2} = \frac{\ln 2}{k} = \frac{15\ln 2}{-\ln(0.80)} \approx \frac{15(0.6931)}{0.2231} \approx 46.6$ days.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Exact values with sum and difference formulas

Find the 4th term of the geometric sequence with $a_1 = 2$ and $r = 3$.

a. **54** ✓ **b.** 162

c. 56 **d.** 57

Applying Sequences principles, $a_4 = 2(3)^3 = 54$. Therefore, the verified correct option is 54.

22 [Foundational] • Target: Working with given values

Find the 3rd term of the geometric sequence with $a_1 = 4$ and $r = 3$.

a. 108 **b.** 40

c. **36** ✓ **d.** 41

Applying Sequences principles, $a_3 = 4(3)^2 = 36$. Therefore, the verified correct option is 36.

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23 [Foundational] • Target: Verifying identities using sum and difference formulas

In the standard (x, y) coordinate plane, the system of equations $\{y = |x^2 - 4|; y = 2x + k$ has exactly 3 distinct real solutions (x, y). What is the sum of all possible values of the real constant k?

a. 4

b. 5

c. **9** ✓

d. 11

The system represents the intersections of the absolute-value quadratic graph $y = |x^2 - 4|$ and the family of parallel lines $y = 2x + k$ of slope 2. The piecewise definition of $y = |x^2 - 4|$ is: \[y = \begin{cases} x^2 - 4, & \text{if } |x| \ge 2 \\ 4 - x^2, & \text{if } |x| 2 \) and $1 - \sqrt{10}5$: 2 solutions (outer branches only). - For $4 < k < 5$: 4 solutions (2 on the inverted arch, 2 on the outer branches). - For $-4 < k < 4$: 2 solutions. - For $k = -4$: 1 solution (passing through $(2, 0)$). - For.

24 [Foundational] • Target: Applications

Find the term containing x^5y^1 in the expansion of $(2x + 3y)^6$.

a. $576x^5y^1$ ✓

b. $576x^1y^5$

c. $6x^5y^1$

d. $1152x^5y^1$

The term is $\binom{6}{1}(2x)^5(3y)^1 = \binom{6}{1} \cdot 2^5 \cdot 3^1x^5y^1 = 576x^5y^1$.

25 [Foundational] • Target: Additional practice

A hyperbola has vertices at $(\pm 3, 0)$ and foci at $(\pm 5, 0)$. Find b^2.

a. -16

b. 34

c. **16** ✓

d. 22

Applying Conics principles, $b^2 = c^2 - a^2 = 25 - 9 = 16$. Therefore, the verified correct option is 16.

26 [Foundational] • Target: A worked derivation

Find the asymptotes of the hyperbola $\frac{x^2}{25} - \frac{y^2}{9} = 1$.

a. $y = \pm\frac{5}{3}x$

b. $y = \pm\frac{3}{5}x$ ✓

c. $y = \pm 15x$

d. $y = \pm\frac{8}{5}x$

For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x = \pm\frac{3}{5}x$.

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27 [Foundational] • Target: Sum and difference formulas with tables of values

What is the coefficient of the term containing x^3 in the expansion of $(1+x)^4$?

a. 8

b. 6

c. 1

d. **4** ✓

By the Binomial Theorem, the coefficient of x^3 in $(1+x)^4$ is $\binom{4}{3} = 4$.

28 [Foundational] • Target: Expanding a general sinusoidal expression

Find the foci of the hyperbola $\frac{x^2}{25} - \frac{y^2}{9} = 1$.

a. $(\pm 5, 0)$

b. $(\pm\sqrt{34}, 0)$ ✓

c. $(\pm 2, 0)$

d. $(0, \pm\sqrt{34})$

For a hyperbola, $c^2 = a^2 + b^2 = 25 + 9 = 34$, so $c = \sqrt{34}$. Foci at $(\pm\sqrt{34}, 0)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A geometric interpretation

Find the end behavior of $f(x) = 3x^4 - x$ as $x \to +\infty$.

a. $f(x) \to 0$

b. $f(x)$ oscillates

c. $f(x) \to +\infty$ ✓

d. $f(x) \to -\infty$

Leading term dominates: $f(x) \to +\infty$ as $x \to +\infty$.

30 [Intermediate] • Target: Solving equations using sum and difference formulas

How many petals does the rose $r = \cos(2\theta)$ have?

a. **4** ✓

b. 6

c. 2

d. 5

For $r = \cos(n\theta)$, $n = 2$ is even, so the number of petals is 2n $= 4$.

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31 [Intermediate] • Target: Exact values with sum and difference formulas

Vector $\mathbf{u} = \langle 2, -3 \rangle$ is perpendicular to $\mathbf{v} = \langle 6, k \rangle$. Find k.

a. 6

b. -4

c. -3

d. **4** ✓

$\mathbf{u} \cdot \mathbf{v} = 0 \Rightarrow 2(6) - 3k = 0 \Rightarrow k = 4.$

32 [Intermediate] • Target: Working with given values

Are $\mathbf{u} = \langle 2, 3 \rangle$ and $\mathbf{v} = \langle -3, 2 \rangle$ orthogonal, parallel, or neither?

a. Cannot be determined

b. **Orthogonal** ✓

c. Neither

d. Parallel

$\mathbf{u} \cdot \mathbf{v} = 2(-3) + 3(2) = 0 = 0 \Rightarrow$ orthogonal.

33 [Intermediate] • Target: Verifying identities using sum and difference formulas

Vector $\mathbf{v} = \langle 6, k \rangle$ is parallel to $\mathbf{u} = \langle 3, 4 \rangle$, with $\mathbf{v} = 2\mathbf{u}$. Find k.

a. -8

b. 6

c. **8** ✓

d. 10

$\mathbf{v} = 2\mathbf{u} = 2\langle 3, 4 \rangle \Rightarrow k = 2(4) = 8.$

34 [Intermediate] • Target: Applications

Find the eccentricity of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

a. $\frac{9}{10}$

b. $\frac{3}{5}$

c. $\frac{4}{5}$ ✓

d. $\frac{5}{4}$

$c = \sqrt{a^2 - b^2} = \sqrt{25 - 9} = 4; e = \frac{c}{a} = \frac{4}{5}.$

35 [Intermediate] • Target: Additional practice

Find the radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ by completing the square.

a. 4

b. **5** ✓

c. 25

d. 6

Completing the square: $(x - (2))^2 + (y - (-3))^2 = 25$, so $r = 5$.

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36 [Intermediate] • Target: A worked derivation

For the parabola $y^2 = 8x$, what is the focus?

a. (4, 0)

b. (0, 2)

c. (-2, 0)

d. **(2, 0) ✓**

$y^2 = 4px$ with $4p = 8 \Rightarrow p = 2$; focus $(p, 0) = (2, 0)$.

37 [Intermediate] • Target: Sum and difference formulas with tables of values

Identify the conic section: $y = x^2 + 3$. (form 8)

a. Ellipse

b. Hyperbola

c. Circle

d. **Parabola ✓**

This equation matches the standard form of a parabola.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Expanding a general sinusoidal expression • Cross-Lesson: 9.02 Geometric Sequences and Series

For the limaçon $r = 1 + 1\cos\theta$, what is the maximum value of r?

a. 0

b. 3

c. 1

d. **2 ✓**

Maximum r occurs at $\cos\theta = 1$: $r_{max} = 1 + 1 = 2$.

39 [Challenging] • Target: A geometric interpretation • Cross-Lesson: 9.03 The Binomial Theorem

A sequence is defined by $a_1 = 5$ and $a_n = a_{n-1} + 3$ for $n \geq 2$. Find a_6.

a. **20 ✓**

b. 18

c. 17

d. 23

This is arithmetic with $d = 3$: $a_6 = 5 + (6 - 1)(3) = 20$.

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40 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 1.01 Functions and Function Notation

A vector $\mathbf{u} = \langle 3, k \rangle$ has magnitude 5. Find the positive value of k, given $k = 4$ is one solution.

a. **4** ✓

b. 2

c. 3

d. 5

$\sqrt{3^2 + k^2} = 5 \Rightarrow k^2 = 16 \Rightarrow k = 4$.

41 [Challenging] • Target: Exact values with sum and difference formulas • Cross-Lesson: 1.02 Domain and Range

Write $\log_2(32) = 5$ in exponential form.

a. $2^5 = 32$ ✓

b. $32^5 = 2$

c. $2^{32} = 5$

d. $5^2 = 32$

$\log_b(x) = y \Leftrightarrow b^y = x$, so $2^5 = 32$.

42 [Challenging] • Target: Working with given values • Cross-Lesson: 1.03 Average Rate of Change

What is the vertex of the parabola $y = (x - 3)^2 + 4$?

a. **(3, 4)** ✓

b. (3, -4)

c. (-3, -4)

d. (-3, 4)

For $y = a(x - h)^2 + k$, the vertex is $(h, k) = (3, 4)$.

43 [Challenging] • Target: Verifying identities using sum and difference formulas • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Find the remainder when $f(x) = x^3 - 2x + 1$ is divided by $x - 2$.

a. **5** ✓

b. 7

c. 2

d. -5

By the Remainder Theorem, remainder $= f(2) = 5$.

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44 [Challenging] • Target: Applications • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the degree and leading coefficient of $f(x) = 3x^4 - 2x^2 + 1$.

a. degree 4, leading coeff -3

b. degree 4, leading coeff 3 ✓

c. degree 3, leading coeff 3

d. degree 5, leading coeff 3

Highest power is 4, its coefficient is 3.

45 [Challenging] • Target: Additional practice • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Convert $\frac{5\pi}{6}$ to degrees.

a. 100°

b. 120°

c. **150°** ✓

d. 180°

$\frac{5\pi}{6} \times \frac{180}{\pi} = 150°$

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